



Strathmore
UNIVERSITY

School of Computing and Engineering Sciences
Bachelor of Science in Computer Networks and Cybersecurity
Semester 1 (April 2025– July 2025)

Course Outline- 2025

Unit Code: CNS 2104

Unit Name: Data Network Design and Management 1

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Purpose of the Unit

To provide students with knowledge and skills in the design and implementation of data networks.

Overall Learning Outcomes

At the end of this course, the student will be able to:

- i. Demonstrate and apply networking principles.
- ii. Apply traffic modelling skills in design of packet switched data networks.
- iii. Apply gained skills for network design in a heterogeneous environment.

Contact Hours

60 Hours

Pre-Requisites

CNS1203– Introduction to Computer Networks

Week	Topic	Intended Learning Outcome	Activities
1-2	- Course Outline - Review of Addressing-IPv4 and IPv6	a. Semester Expectations b. Review IPv4 addresses c. Review of IPv6 addresses	Addressing Lab and Discussion Q & A
3	DHCP Concepts - DHCP IP address allocation mechanisms - DHCP discover messages - DHCP client/server interaction	a. Differentiate DHCP IP address allocation mechanisms b. Name and describe DHCP discover messages c. Explain DHCP client/server interaction d. Explain DHCP for IPv6	Lecture, Discussions, Student research, Problem solving Assign: DHCPv6
4	Intro to Switching and Routing - Parts of a Switch, Switch MAC Table - Parts of a router, Router as a computer - Routing table concepts-structure, population, search - Static routes; types and implementation	a. Physically describe the internal parts of a Switch b. Switching Table entries c. Physically describe the internal parts of a router, role of the default gateway d. Identify and explaining routing table entries e. Explain different types of static routes and their implementation.	Lecture, Discussions, Student research, Problem solving, Packet Tracer configuration of switches and Routers Lab – Static Routing
5	Network/LAN Design Two-Tier (Collapsed Core) Architecture	a. Discuss the components of two-tier and three-tier switched LAN designs b. Explain the principles of the switched	Lecture, class discussion, student research, Lab - Lab Assignment - Simulation of

	• Three-Tier Architectures	LAN architecture Simulate a simple three-tier LAN architecture applying principles learned	a three-tier switched LAN design
6-7	Switch Concepts • Types of Switches: Managed (Enterprise, Smart), PoE, LAN • Switch Configuration and Management • Basic Switch Security Configuration • Switch Architecture Switch Domains (Collision, Broadcast)	a. Differentiate between various switch frame forwarding methods b. Describe how the CAM table is built and searched c. Differentiate between collision and broadcast domains d. Describe various types of switches Configure and troubleshoot managed switches	Lecture, class discussion, student research, Lab - Lab- Configure basic switch settings and examine CAM tables Lab-Configure and troubleshoot basic switch security settings
8-9	VLANs and Inter-VLAN routing • VLAN Trunk Operation • Inter-Vlan Routing Strategies (Router-On-A-Stick, SVI, Per-Interface) • Implementing and Troubleshooting Inter-VLAN Routing	a. Understand switching principles. b. VLAN types and meaning c. VLAN creation and management d. VLAN trunking e. VTP for VLAN management	Discussions VLAN configuration Labs Lab -Design, configure, manage and troubleshoot VLANs Lab - configure and troubleshoot inter-VLAN routing Study/ Configure 1: VTP
10	STP • Role of STP in switched LANs • STP components and Operation	a. Review of STP b. MSTP and RSTP c. Inter-VLAN routing ○ Router on a stick ○ SVI and per interface routing	Cases and Journal Paper reviews Lab - Simulate VTP configuration based on reading assignment Reading Assignment 2: PVST+

	STP Implementation: CST, RSTP MSTP Etc.		
11	EtherChannel Port Aggregation Protocol (PAgP) operation Link Aggregation Control Protocol (LACP) ops	a.Explain the role of EtherChannel in switched LANs b. Discuss PAgP mode settings c.Discuss LACP mode settings	Lectures, Class Discussion, Lab-PAgP & LACP labs
12	Wireless LAN - Introduction - Standards, Components and Operation - Wireless LAN Planning - WLAN Security- threats, protocols WLAN Implementation and Troubleshooting	a.Discuss various WLAN standards b. Describe WLAN infrastructure c.Describe how WLAN s operate d. Discuss the WLAN planning process e.Discuss the threats and protocols used to secure a WLAN Simulate a WLAN implementation	Lecture, Class discussions, Student research, Problem solving, Lab – WLAN

Assessment Breakdown

Unit Assessment Type	Weighting (%)
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Final examination	50
CAT 1	10
CAT 2	10
Lab Assignments	10
Group Presentations	10
Total	100

Mode of Delivery

- Lectures, Tutorials and Labs

Core Reading Materials

1. CCNAv7 Enterprise Networking, Security, and Automation
2. Computer Networking: A Top-Down Approach (7th Edition), James Kurose and Keith Ross, Pearson, 7th ed. (2016), ISBN-10: 0133594149 (ISBN-13: 978-0133594140)
3. Data Communications and Networking, Behrouz Forouzan, McGraw-Hill, 5th ed. (2012), ISBN-10: 0073376221 | ISBN-13: 978-0073376226

Recommended Reference Materials

1. Communications Systems and Networks, Ray Horak, Wiley; 3rd ed., 2002, ISBN-10: 0764548999 (ISBN-13: 978-0764548994)
2. Network Analysis, Architecture and Design, McCabe, J. D., Morgan-Kaufmann (2003), 1558608877, 978-1558608877
3. Wireless Communications & Networks, William Stallings, Prentice Hall, 2nd ed. (2004) ISBN-10: 0131918354 ISBN-13: 978-0131918351
4. Data and Computer Communications, W. Stallings, Prentice Hall, 10th ed. (2013), ISBN-10: 0133506487 ISBN-13: 978-0133506488

E-resources

1. Cisco Networking Academy-www.netacad.com

Lectures/Classes

1. Punctuality is **Important**
2. Active participation in class discussions is encouraged

Assignments and/or Coursework

1. **Plagiarism** is a serious offence. If detected in any form the following will apply:
 - a. In partial cases (such as not citing whole word-for-word quotes), half the total possible marks of the assignment shall be duly struck off
 - b. In serious cases (such as whole duplication of an assignment), a zero policy will apply, i.e., all offending assignments shall be awarded a mark of zero.

Note: The level of seriousness referred to above is at the discretion of the lecturer. Appeals are certainly possible through the relevant channels.
2. Notwithstanding the above, **collaboration** in coursework is encouraged as this promotes team spirit and group synergy (as long as originality is preserved). If you consult other references for an assignment, you must cite them using the 6th edition of the APA style.

Communication Channels

1. e-mail (**Maintain formality**)
2. Unit leader- ahmed.asad@strathmore.edu ; prtipal.kaur@strathmore.edu
 - a. Group A-Ramla Dimbil; Okwudili Anthony
 - b. Group B- Abraham Mugwe and Umutoni Bucagu
3. Google Classroom platform- <https://classroom.google.com/c/NzY1MjI5NTQ0Njgx?cjc=zeuaiph6>